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Continuum of denitrification to end-member controls on nitrate isotopic ratios: evidence from meta-analyses and a six-year record of seasonality from Kentucky, USA

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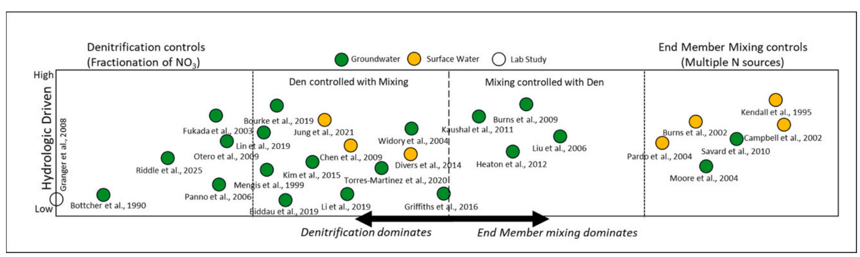
HIGHLIGHTS

- Empirical mode decomposition show seasonality of nitrate isotopic ratios across six years
- Seasonality of nitrate isotopic ratios is controlled by soil nitrogen and denitrification
- Seasonal variation explains widely reported denitrification line and Rayleigh diagram
- Continuum of denitrification to end-member controls shown across prior studies
- Multiple sources may impart equifinality in numerical modelling

GRAPHICAL ABSTRACT



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ABSTRACT

The isotopic ratios of nitrate ($\delta^{15}N_{NO3}$ and $\delta^{18}O_{NO3}$) are common tracers of nitrate source and transformation in soil and aquatic systems. However, investigation of their seasonality is limited owing to seldom reported multi-year datasets. This study collects groundwater and springhead samples, measures and analyzes a six-year dataset from the inner bluegrass karst region of central Kentucky, USA to analyze seasonal patterns of nitrate isotopic ratios and investigate their controls. We observe distinct isotopic measurements in groundwater and at the springhead during the wetter winter-spring season ($\delta^{15}N_{NO3} = 6.0 \pm 1.6‰$ and $\delta^{18}O_{NO3} = 1.6 \pm 2.3‰$) compared to the drier fall season ($\delta^{15}N_{NO3} = 9.0 \pm 1.1‰$ and $\delta^{18}O_{NO3} = 3.4 \pm 1.7‰$). Results of time series analyses and empirical mode decomposition suggest the seasonality of the isotopic ratios in the aquifer are controlled by intra-annual variability of soil nitrate isotopic composition, with leaching of nitrate mineralized from soil nitrogen dominating the winter signal and partially denitrified nitrate dominating the fall signal.

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Results show seasonality of the denitrification line and seasonality of the Rayleigh diagram attributed to the dominant soil nitrogen source and the denitrification controls. Meta-analysis of data from 28 journal papers reporting $\delta^{15}N_{NO3}$ and $\delta^{18}O_{NO3}$ measurements of groundwater and streamwater show a continuum of denitrification-controlled to end-member-controlled results. Groundwater and surface water studies with a dominant end-member of nitrate show potential estimating denitrification rates with assistance from $\delta^{15}N_{NO3}$ and $\delta^{18}O_{NO3}$ as tracers. Studies with water exhibiting relatively fast transport from multiple nitrate sources show potential for $\delta^{15}N_{NO3}$ and $\delta^{18}O_{NO3}$ as tracers in end-member mixing. Researchers should exercise caution when multiple sources exhibit some denitrification influence, as equifinality will be problematic in numerical modelling.

1. Introduction

Measurements of the isotopic ratios of nitrate ($\delta^{15}N_{NO3}$ and $\delta^{18}O_{NO3}$) in groundwater and streamwater can provide insight to nitrate transformation and sources in a basin (Kendall, 1998; Kendall and Aravena, 2000). In turn, the challenge using the isotopic ratios as tracers lies in differentiating between denitrification and end-member influence as variance is influenced by both factors (see Fig. 1). Recent work has shown how $\delta^{15}N_{NO3}$ versus $\delta^{18}O_{NO3}$ plots can be used to identify denitrification impacted systems (Boumaiza et al., 2024) and review using N and O isotope fractionation for evaluating denitrification (Margalef-Marti et al., 2025).

Karst groundwater systems, particularly those associated with agricultural land use, are notably under-studied in terms of long term nitrate isotopic ratio datasets. Karst systems include the presence of fracture networks, conduits, and caves that provide fast transit of karst groundwater and its nitrate (Husic et al., 2021). These unique hydrogeological characteristics of karst systems, including these multiple flow pathways, provide low residence time in groundwater such that source to basin outlet transport may be on the order of hours to weeks (Al Aamery et al., 2021), as opposed to years or decades. The fast groundwater transit in karst provides an opportunity to investigate an integrated signal of nitrate isotopic ratios with low probable occurrence of nitrate transformation from source to basin outlet. Therefore, we hypothesized that a relatively long 6-years duration dataset of $\delta^{15}N_{NO3}$ and $\delta^{18}O_{NO3}$ from karst groundwater would provide measurements from nitrate sources in the karst basin, as well as allow analyses of their seasonality.

There remains a significant lack of research connecting the seasonality of nitrate data to cross-basin source and transformation processes. The existing literature reveals insufficient long-term time series data that can effectively link the seasonality of nitrate concentrations and isotopic ratios to source and denitrification processes (Pina-Ochoa and Álvarez-Cobelas, 2006). Most published studies carry out sampling of water containing nitrate and analyzed for $\delta^{15}N_{NO3}$ and $\delta^{18}O_{NO3}$ over less than two years, and often less than one year (Table 1, column 5); and the frequency and number of samples often fall short of capturing discernible seasonal patterns. Additionally, many authors have focused on specific flow regimes, such as low flow versus storm pulses, without adequately addressing the seasonal dynamics of nitrate isotopic ratio data. This paper aims to address these gaps by analyzing the seasonality of the nitrate data over a six-year duration, such that repetition of dry and wet hydrologic regimes might provide sufficient information to infer controls on $\delta^{15}N_{NO3}$ and $\delta^{18}O_{NO3}$.

As part of this study, we consider typically reported metrics of nitrate dentification inferred using $\delta^{15}N_{NO3}$ and $\delta^{18}O_{NO3}$, including the denitrification line gradient and Rayleigh diagrams. Recent studies have reported on the denitrification line and reviewed isotopic effects during denitrification (Boumaiza et al., 2024; Margalef-Marti et al., 2025) and others highlight its significance in elucidating the relationship between nitrate concentrations and isotopic ratios during denitrification processes (Böttcher et al., 1990; Mengis et al., 1999; Fukada et al., 2003; Widory et al., 2004; Panno et al., 2006; Chen et al., 2009; Divers et al., 2014; Kim et al., 2015; Griffiths et al., 2016; Bourke et al., 2019; Li et al., 2019; Lin et al., 2019; Biddau et al., 2019; Torres-Martínez et al., 2020; Husic et al., 2020a; Jung et al., 2021). These studies emphasize the utility of the denitrification line in tracking nitrogen transformations and assessing the efficiency of denitrification in various environments. Similarly, the Rayleigh diagram was recently applied in numerous

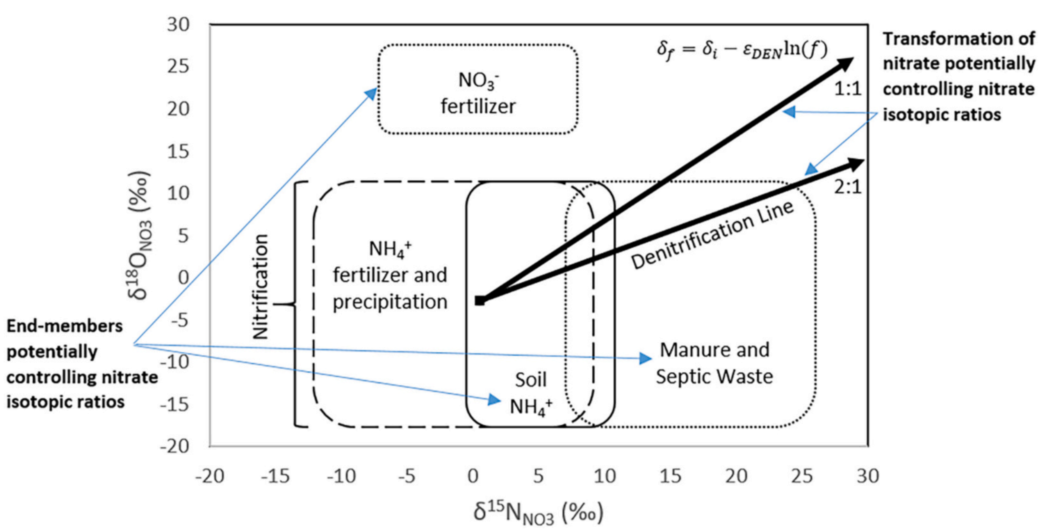

Fig. 1. Controls on $\delta^{15}N$ and $\delta^{18}O$ of nitrate in surface water and groundwater, including (1) End-members potentially controlling nitrate isotopic ratios and (2) Transformation of nitrate potentially controlling nitrate isotopic ratios. The typical values of $\delta^{15}N$ and $\delta^{18}O$ of nitrate sources are derived from the literature, after Kendall and Aravena (2000). The two arrows indicate typical expected slopes for data resulting from denitrification of nitrate with initial $\delta^{15}N = 0‰$ and $\delta^{18}O = -2‰$. The typical ranges of $\delta^{18}O_{NO3}$ values produced by nitrification of ammonium and organic matter denoted by "nitrification" as presented by Kendall and Aravena (2000).

Table 1
Nitrate isotopic ratios ($\delta^{15}N$, $\delta^{18}O$) applied for analyzing nitrate transformation and sources in surface and groundwater studies.

Study	Location	Watershed (km^2)	System Type	Sampling Duration (months)	Land Use	Isotope Time Series	Denitrification Line ($\delta^{18}O$ vs. $\delta^{15}N$)	Vector Slope	Rayleigh Plot ($\delta^{15}N$ vs. fNO_3)
Böttcher et al., 1990	Fuhrberger Field aquifer, Hanover, Germany	300*	Unconfined shallow groundwater	–	Arable land, coniferous forest, and low intensive grassland	No	Yes	**0.48 (2.1:1)** *0.48 (2.1:1)*	Yes
Kendall et al., 1995	Catskill Mountains, USA; New York, Colorado, Vermont	0.47*	Upland catchment with surface river	3	Forested	No	Yes		No
Mengis et al., 1999	Waterloo, Ontanrio, Canada	2.7	Groundwater riparian zone	1	Primarily agriculture corn, soybeans, winter wheat, and strawberries	No	Yes	*0.67 (1.5:1)*	No
Burns and Kendall, 2002	Neversink River, Catskill Mountains, New York, USA	16, 96	Upland catchment with surface river	18	Completely forested	Yes	No		No
Campbell et al., 2002	Loch Vale watershed, Colorado, USA	18.3, 32.6	Alpine (upland) catchment with surface river	36	Primarily bare rock, boulder fields, snow, and ice with tundra and forest	Yes	Yes	–	No
Fukada et al., 2003	Torgua, eastern Germany	36	Groundwater river-bank infiltration	1	Primarily grassland for sheep pasture and arable agriculture	No	Yes	*0.76 (1.3:1)*	Yes
Widory et al., 2004	Arguenon watershed, Brittany, France	0.33, 1.25	Groundwater	16	Primarily agriculture with pig farming, poultry and cattle breeding	No	No		Yes
Pardo et al., 2004	White Mountains, New Hampsire, USA		River	20	Disturbed and undisturbed forests	Yes	Yes		No
Moore et al., 2006	Livermore Valley GW Basin, California, USA	282*	River and Groundwater	12	Agriculture with livestock farming and vineyards, urban development	No	No		No
Panno et al., 2006	Mississippi River, Illinois, USA	4.76 million*	River and tile drain	9	Primarily row-crop agriculture in corn belt	No	Yes	*0.59 (1.7:1)*	No
Liu et al., 2006	Karst groundwater of Guiyang, China		River and Groundwater	2	Primarily agriculture	No	Yes		No
Otero et al., 2009	Plana de Vic, Osona, Spain	600	Surface and Groundwater	3	Primarily agriculture	No	Yes	*0.55 (1.8:1)*	Yes
Chen et al., 2009	Beijing River, south China	46,710	Surface River	9	Agriculture, primarily rice cultivation	No	Yes	**0.48 (2.1:1)** *0.44 (2.3:1)*	Yes
Burns et al., 2009	Many sites in New York, USA		Surface and Groundwater	15	Mix of agriculture, urban and forested watersheds	No	Yes	**0.50 (2.0:1)**	Yes
Savard et al., 2010	Wilmot River Basin, PEI, Canada	87	Surface and Groundwater	24	Agriculture with some forests, intensive potato production	No	Yes		No
Kaushal et al., 2011	Many watersheds in Batimore, MD, USA		Surface and Groundwater	6	Forested and agricultural watersheds, developing watersheds	No	Yes	**0.50 (2.0:1)**	No
Heaton et al., 2012	Republic of Malta, Malta and Gozo aquifers		Surface and Groundwater	3	Agriculture and urban areas with sewer galleries and livestock enclosures	No	Yes		No
Kim et al., 2015	Haean groundwater basin, South Korea	64	Surface and Groundwater	1	Forested, agriculture (rice paddy fields), residential area	No	Yes	*0.35 (2.9:1)*	No
Divers et al., 2014	Nine Mile Run, Pittsburgh, Pennsylavania, USA	15.7	Stream	20	Urban with some impervious cover	No	Yes	**0.43 (2.3:1)** *0.51 (2.0:1)*	No

(continued on next page)

studies for visualizing isotopic fractionation during denitrification (Böttcher et al., 1990; Fryar et al., 2000; Fukada et al., 2003; Widory et al., 2004; Chen et al., 2009; Griffiths et al., 2016; Bourke et al., 2019; Biddau et al., 2019; Li et al., 2019; Jung et al., 2021). We investigate these metrics for interpreting isotopic data and understanding the underlying processes that govern nitrogen cycling in different ecosystems.

As a final focus of investigation, we carried out a meta-analysis of controls on the nitrate isotopic ratios for datasets published in 28 journals articles the past 35 years to allow discussion of the modelling application that uses $\delta^{15}N_{NO3}$ and $\delta^{18}O_{NO3}$ as tracers. The challenge of tracer-aided numerical modelling with nitrate isotopic ratios lies in differentiating between transformation processes and nitrogen source mixing, as nitrate isotopic ratio variance is influenced by both factors (see Fig. 1). This presents a significant knowledge gap in the field of nitrogen modelling, particularly in understanding how various systems indicate transformations versus nitrogen source mixing. By leveraging nitrate isotopic ratio data, we aim to discuss the calibration of numerical models, as the tracers provide an extra set of equations with a limited number of unknowns, provided these unknowns are not overly sensitive or can be easily constrained. Previous research has indicated that isotopic ratios of nitrate can serve as effective tracers in modelling efforts (e.g., Ford et al., 2017; Li et al., 2019; Torres-Martínez et al., 2020; Husic et al., 2020a). However, the advancement of tracer-aided numerical modelling with the isotopic ratios remains hindered by the complexities of distinguishing between transformation processes and nitrogen source mixing. This paper seeks to fill that knowledge gap, by providing a meta-analysis of existing literature and identifying the controls on $\delta^{15}N_{NO3}$ and $\delta^{18}O_{NO3}$ in these reported studies.

The objective of this study was to analyze the seasonal patterns of nitrate isotopic ratios in the inner Bluegrass Region of central Kentucky, emphasizing the transformations and end member mixing that influence these patterns (Fig. 2). Utilizing a six-year dataset, the research aims to elucidate the controls on nitrate concentrations and isotopic ratios across different seasons in karst. Additionally, the study incorporates a meta-analysis of existing literature on nitrate isotopic ratios in surface and groundwater, highlighting how denitrification and source mixing interact to shape isotopic signatures.

2. Study site and materials

The study site is the Cane Run watershed and the Royal Spring groundwater basin, located in the Inner Bluegrass Region of Kentucky, USA (Fig. 3). This basin covers an area of approximately 96 km^2, with the groundwater basin itself extending over 58 km^2. The region is characterized by a temperate climate, with an average annual temperature of 13.0 ± 0.7 °C and a mean annual precipitation of 1170 ± 200 mm. The topography features rolling hills and mild relief, which are conducive to agricultural practices, particularly horse farming (Paylor and Currens, 2004). The underlying geology consists of the Lexington Limestone from the Middle Ordovician period, which is highly karstic and well-developed, contributing to the unique hydrological features of the area. Human disturbances, primarily from agricultural practices, have altered land cover, with approximately 60% of the basin being agricultural land and 40% urbanized (UKCAFE, 2013).

The features of the Cane Run watershed that are particularly relevant to the study's focus regarding nitrogen transport and biogeochemical

Table 1 (*continued*)

Study	Location	Watershed (km^2)	System Type	Sampling Duration (months)	Land Use	Isotope Time Series	Denitrification Line (δ^{18}O vs. δ^{15}N)	Vector Slope	Rayleigh Plot (δ^{15}N vs. fNO$_3$)
Griffiths et al., 2016	New Ellenton, South Carolina, USA		Surface and Groundwater	12	Reforested from row crop agriculture in 1951	No	Yes	*1.09 (0.9:1)*	Yes
Biddau et al., 2019	Arborea plain, Sardinia Italy	60	Surface and Groundwater	12	Agriculture and livestock	Yes	Yes	**0.40 (2.5:1)** *0.34 (2.9:1)*	Yes
Lin et al., 2019	Upper Illinois River Basin, USA	37,353	River and Groundwater	48	Primarily consisting of agricultural and urban areas	No	Yes	*0.55 (1.8:1)* *0.46 (2.2:1)*	No
Li et al., 2019	Xiajang River, China	353,120	River	12	Forest and cultivated land, with urbanized and industrial activities	Yes	Yes	**0.59 (1.7:1)** *0.71 (1.4:1)*	Yes
Bourke et al., 2019	Alberta, Canada		Groundwater	3	Agriculture and livestock	No	Yes	*0.72 (1.4:1)* *0.42 (2.4:1)*	Yes
Torres-Martínez et al., 2020	Monterrey valley, Mexico	6680	Surface and Groundwater	1	Cropland and urban area	No	Yes	*0.78 (1.3:1)*	No
Husic et al., 2020b	Royal Springs GW basin and Cane Run watershed, Kentucky, USA	58, 96	Stream and Groundwater	2	Primarily agriculture with urbanized head waters	Yes	Yes		No
Jung et al., 2021	WWTP and Bogang stream, Jeungpyeong, South Korea	158	Stream	9	Agriculture and urban residential	Yes	Yes	*0.52 (1.9:1)* *0.32 (3.1:1)*	Yes
This Study, 2025	Royal Springs GW basin and Cane Run watershed, Kentucky, USA	58, 96	Stream and Groundwater	72	Primarily agriculture with urbanized head waters	Yes	Yes	**0.83 (1.2:1)** *0.83 (1.2:1)*	Yes

Bold vector slopes are reported by authors in each study and italicized vector slopes are calculated from data presented in Fig. 10.

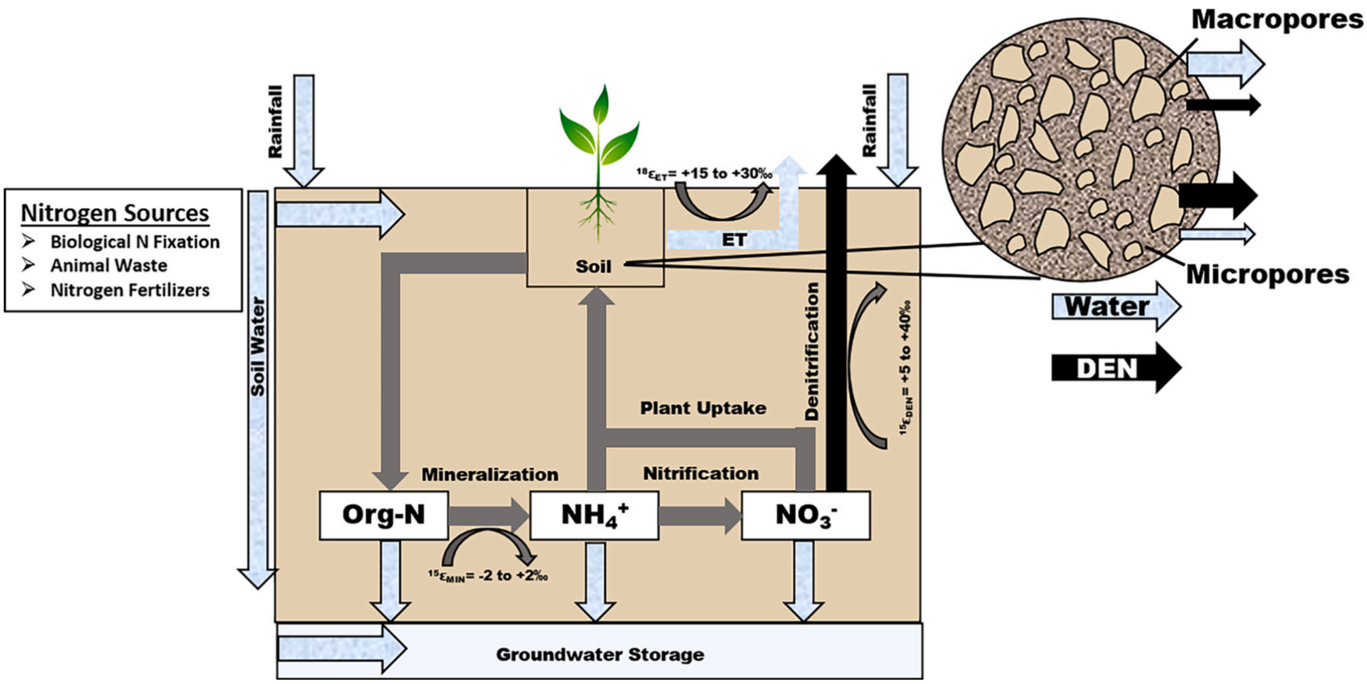

Fig. 2. Water, nitrogen and nitrogen isotopic ratio dynamics in soil-groundwater systems. Concept model demonstrating how water and nitrogen move through soil and groundwater systems. Precipitation infiltrates both vertically and horizontally, eventually draining through soil leaching. The soil's pore spaces contain water, which serves as a medium for nutrients and microbial activity affecting nitrogen transformation. Water loss occurs through evapotranspiration (ET), combining soil evaporation and plant transpiration. Plants extract nutrients for growth, while microorganisms facilitate nitrogen cycling: organic nitrogen transforms to ammonium, then to nitrate through nitrification, and finally to gaseous nitrogen through denitrification. These transformations follow Rayleigh-like isotope fractionation. Hypothesized processes include soil pore structure significantly influences water retention and microbial processes: micropores support pronounced water retention due to surface tension, creating oxygen-poor conditions ideal for denitrifying bacteria; and macropores allow rapid water movement and better aeration, supporting aerobic processes like mineralization and nitrification.

processes within karst systems include the following: the presence of numerous karst features, including over 50 sinkholes and swallets, facilitates rapid transport of water and nutrients from surface to subsurface environments (Taylor, 1992; Paylor and Currens, 2004). 40% of the watershed has shallow surface slopes (0–3%); 50% has moderate slopes (3–6%) and 10% has steeper slopes (>6%) (Al AAmery et al., 2021). Additionally, the low to mild gradients in the watershed provide soil pore sites for water and nitrogen retention during periods of lower soil moisture content such as drier summer and fall months. The soils in this region are primarily silt loam, which influences the infiltration and retention of water and nutrients, thereby affecting the overall hydrology of the watershed. The fast transport dynamics observed in the phreatic karst cave suggest that sediment and nutrient fluxes can occur rapidly, especially during storm events, aligning with previous study's focus on the timing and pathways of nitrogen export (Husic et al., 2020a). Soil characteristics and hydrological processes support microbial processes that influence nitrate concentrations and isotopic ratios seasonal variance, and highlight the importance of examining the biogeochemical cycling of nitrogen in this karst landscape.

Sampling locations within the study site were strategically chosen to capture the hydrological and biogeochemical dynamics of the watershed (see Fig. 3). Key nodes include the Groundwater Station at Kentucky Horse Park, which directly intersects the primary phreatic cave, and the Royal Spring, where water and sediment outputs are monitored (Husic et al., 2020a; Bettel et al., 2022). Water depth measurements to the groundwater table vary from 7.4 m to 15 m, with the mean equaling 12.2 m ±1.4 m at the well. The base of the well is situated at 240 above mean seal level and the land surface at 258 above mean sea level. Subsurface flows are routed to a phreatic cave that supplies the primary basin outlet (Royal Spring, 243 m above sea level) with an average perennial discharge of 0.67 m^3/s (Husic et al., 2020a). Additional sampling stations are established at tributaries upstream of swallets to assess nutrient contributions from both urban and agricultural land uses. These locations facilitate the collection of water and nutrient samples during various flow conditions, including storm events and low-flow periods, allowing for analysis of nutrient dynamics and transport mechanisms within the karst system. This approach allowed investigating the seasonal patterns of nitrate isotopic ratios and their correlation with hydrological variables.

Materials included publicly available datasets published by government agencies. The United States Geological Survey (USGS) operates a v-notch weir at Royal Spring (USGS 03288110) and provided flowrate data for the Royal Springs sampling site. Web Soil Survey (WSS) provided soil data and information published by the USDA Natural Resources Conservation Service (NRCS). Soil moisture content was collected and published by Kentucky Mesonet at the Versailles, KY sampling.

3. Methods

3.1. Sampling and analyses for nitrate concentration and nitrate isotopic ratios

At the well at Kentucky Horse Park and at Royal Springs, samples were collected in bulk 1 L bottles and analyzed for NO_3^-, $\delta^{15}N_{NO3}$ and $\delta^{18}O_{NO3}$. Samples were collected bi-weekly for the six-year period from January 2017 to December 2023.

The samples for analyses of NO_3^- concentration in water were filtered into sterile 125 mL plastic bottles and analyzed with the Kentucky Geological Survey following US EPA Method 300.0. The analysis was performed using a Dionex ICS-3000 Ion Chromatography System featuring a carbonate–bicarbonate eluent generator and Dionex AS4A analytical column. Retention time was used to identify the NO_3^- anion, and peak areas were compared to calibration curves generated from

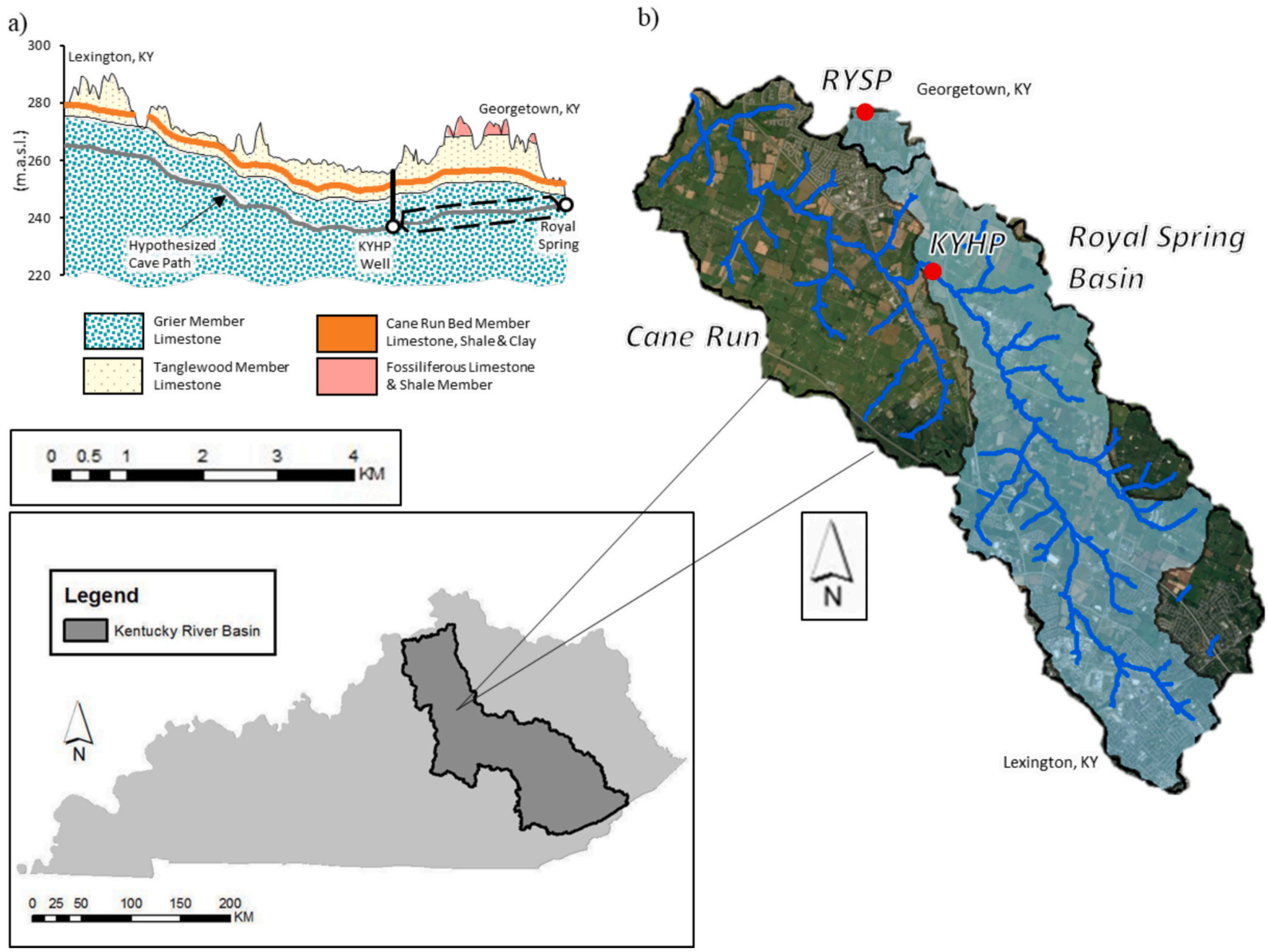

Fig. 3. (a) Profile view of the hypothesized cave path beneath the Cane Run watershed along with the superposition of geologic members. The dashed black box around a portion of the conduit path indicates the study section. (b) Cane Run watershed and Royal Spring basin indicating the location of the Kentucky Horse Park well (cave) and Royal Spring (outlet) sites.

known standards.

Samples for nitrate isotopic ratios were extracted in the lab from the bulk 1 L bottles using clean 60 mL syringes and passed through 0.45 μm syringe filters (Whatman 6780–2504) into sterile 40 mL borosilicate vials (I-Chem TB36–0040). Samples were stored in a refrigerated environment without the use of preservatives before delivery to University of Arkansas Stable Isotope Lab (UASIL) for analysis.

For comparison of nitrate isotopic ratios with that of soil nitrogen, surface soil samples were collected at five locations from soils with hay cover in the inner bluegrass on one occasion in March 2024. Sampling was performed using a garden trowel on the top 0–2 cm soil layer and approximately 30 g of sample were collected. Pristine and disturbed soil sites were identified and pooled within each location to account for inter-site variability. Samples were then split into fifteen 10 g aliquots and oven dried at 60 °C. Samples were stored in a cool, dry environment before delivery to University of Arkansas Stable Isotope Lab (UASIL) for analysis.

At UASIL, stable isotopic ratios of nitrate were measured using a bacterial denitrification method analyzed on a Thermo Gas Bench II interfaced to a Finnigan DeltaPlus IRMS (Sigman et al., 2001; Casciotti et al., 2002). The isotopic reference materials for NO_3^- were USGS32 ($\delta^{15}N_{NO3}$ = + 180‰), USGS34 ($\delta^{15}N_{NO3}$ = −1.8‰, $\delta^{18}O_{NO3}$ = −27.9‰), and USGS35 ($\delta^{18}O_{NO3}$ = + 57.5‰). Average standard deviations for the NO_3^- isotopic standards were 2.03‰ for USGS32 for $\delta^{15}N$; 0.34 and 0.70‰ for USGS34 for $\delta^{15}N$ and $\delta^{18}O$, respectively; and 1.00‰ for USGS35 for $\delta^{18}O$. Duplicates of $\delta^{15}N_{NO3}$ and $\delta^{18}O_{NO3}$ ($n = 5$) had standard deviations of ±0.28‰ and ± 0.45‰, respectively.

For isotopic analyses of soils, powdered soil samples were weighed into tin capsules. TN and $\delta^{15}N$ were analyzed using a Costech 4010 elemental analyzer interfaced with a Thermo Finnigan Conflo III device to a Thermo Finnigan Delta Plus XP isotopic ratio mass spectrometer (IRMS). Average standard deviation for USGS-41 was 0.13‰ for $\delta^{15}N$. Average standard deviation for $\delta^{15}N$ of unknowns was 0.11‰.

All isotopic ratios (δ) are reported in units of per mil (‰), which represents the relative abundance of heavy-to-light isotopes in a sample, and is calculated as follows:

$$\delta = \left(\frac{R_{sample} - R_{standard}}{R_{standard}}\right) \times 1000,$$

where R is the ratio of the abundance of the heavy-to-light isotopes, sample is the field sample, and standard is the reference standard of the known isotope ratio. The references used for the analysis of N and O isotopes were AIR and Standard Mean Ocean Water (SMOW), respectively.

3.2. Data post-processing and time series analyses

Data collected from the two sampling sites were compared and used as sample repetitions in our analyses. Time series over the six years of data collection provided visualization of trends and seasonal patterns of hydrological and nitrate measurements within the study site. Isotope bi-plot (δ^{15}N vs. δ^{18}O) and fractionation bi-plot (δ^{15}N vs. ln(NO_3^-)) provided comparison between two variables; and aided in constructing theoretical development of the denitrification line and Rayleigh diagram that influence nitrate concentrations and isotopic ratios. Seasonal patterns identified as maximum and minimum data measurements were investigated at monthly intervals and grouped in agreement with hydrologic seasonality.

3.3. Empirical mode decomposition

Time series decomposition methods allow quantitative statistical evaluation of the timing and seasonality of nitrate measurements in a basin (Ford et al., 2017; Gerlitz et al., 2023). Empirical mode decomposition (EMD) provided the authors a data-driven approach to assess significance of oscillations at varying time-scales (Huang et al., 1998). EMD provides some advantages over other time-series analysis as the method does not assume parametric, linear or stationary characteristics of datasets (Huang et al., 1998; Wu et al., 2007). Empirical mode decomposition was used to separate the 6-year time series of NO_3-N, δ^{15}N and δ^{18}O to the statistically significant intrinsic mode functions. EMD was performed using MATLAB version R2019b. A statistical significance test was performed on EMD results based on the method explained in Wu et al. (2007) to determine if IMFs were significantly different from noise IMFs. Statistically significant frequencies between two and eighteen months were included in our analysis since we were interested in intra-seasonal and seasonal timing of peaks, effectively removing event-based and longer-term trends.

3.4. Meta-analysis

A total of 27 published nitrate isotopic ratio studies were found to use nitrate isotope data to investigate nitrogen transformations and source mixing in surface and groundwater systems. These papers were reviewed for system type, land use, sampling duration as shown in literature table (see Table 1). Each study was noted for inclusion of isotope time series, denitrification line (i.e. isotope bi-plot), and Rayleigh diagram (i.e. fractionation plot).

Data from the 27 studies was extracted from isotope bi-plots or time series presented in the papers using Plot Digitizer and analyzed to complete our meta-analyses. Data presented in our meta-analyses results only includes surface and groundwater samples, and a breakdown of what data is included is detailed in supplemental information (S2). Nitrate isotopic ratio data was plotted and fit with a linear trend line used to estimate the denitrification line (i.e, slope) for each study. This calculated slope was reported in the literature table, with the denitrification line slope presented by the authors, if available.

The studies included in the meta-analysis were then organized based on an assessment of nitrate isotopic ratios showing control from denitrification (fractionation of N and O isotopic ratios) to end member mixing (multiple nitrogen sources). Studies that indicated fractionation of residual nitrate due to denitrification were classified as were studies dominated by end-member mixing of sources. We qualify many systems will have nitrate isotopic ratios dependent on both denitrification and end member mixing, and we used two classifications to represent studies that show both as important to nitrate isotopic ratios. Studies were also assessed to consider hydrological processes influencing nitrate isotopic ratios.

4. Results and discussion

4.1. Seasonal distribution of nitrate isotopic ratios, and likely controls, for the karst basin

Climate results and hydrologic measurements over the six-year duration show seasonal variation of soil and groundwater hydrology for the karst basin (Fig. 4). Air temperature (Fig. 4a) shows the presence of winter lows and summer high seasonal variation for the humid subtropic region of the southeastern interior portion of the United States, and the region reflects four distinct seasons (winter, spring, summer and fall). Monthly precipitation (Fig. 4a) does not vary pronouncedly, and the region is not inherently wet or dry in terms of rainfall or snowfall occurrence. For example, the lowest average monthly precipitation in November equal to 86 mm does not vary considerably from the highest precipitation in May equal to 138 mm. However, potential evapotranspiration varies considerably across seasons, and the temporal distribution generally matches that of air temperature (Fig. 4b). Monthly average evapotranspiration is nearly an order of magnitude greater in July as compared to January (147 mm in July and 21 mm in January).

The impacts of evapotranspiration variation under fairly consistent monthly precipitation produces relatively wet and dry soil conditions and groundwater regimes for the karst basin (Fig. 4b,c). Soil moisture shows considerable seasonality with wetter soils in winter and spring and drier soils in late summer and fall. February demonstrated the highest soil moisture content, with extreme values reaching saturation and equal to 0.47 m^3/m^3, while late summer and early fall months (August, September, and October) showed the lowest range with soil moisture reaching as low as 0.12 m^3/m^3. On average, winter and early spring (December–March) monthly averages range from 0.34 to 0.38 m^3/m^3; and late summer and fall (July–October) monthly averages range between 0.25 and 0.27 m^3/m^3. Groundwater flowrate from the karst cave at Royal Spring (Fig. 4c) shows seasonality that reflects soil moisture measurements, including pronounced wetter and drier seasons. Royal Spring flowrate showed peak discharge during winter and early spring months (January–March) averaging 1.22 to 1.38 m^3/s; and minimum flow during late summer and fall months (July–October) averaging 0.31 to 0.75 m^3/s. March exhibited the highest flow variability (0.47–3.51 m^3/s), while late summer and early fall months showed the lowest range (0.01–2.08 m^3/s).

Noteworthy in the hydrology measurements is the consistent variation for soil moisture and Royal Spring water flowrate across weekly, monthly and seasonal scales (see Fig. 4c). For example, high precipitation in February 2018 is visually correlated with an increase in soil moisture and Royal Spring flowrate. These observations, as well as the wetter winter and early spring, and drier summer and early fall seasons, observed for soil moisture and spring flow, show the response of groundwater to soil conditions.

Nitrate isotopic ratio measurements and nitrate concentration showed seasonal patterns across the six-year dataset (Fig. 4 d-f). From 2017 to 2020, nitrate concentration measurements showed winter peaks equal to approximately 5 mg/L NO_3-N (January–February) and late summer lows on the order of 2 mg/L NO_3-N (July–August). From 2020 to 2022, similar seasonal patterns persisted, although winter peaks decreased to 3 mg/L NO_3-N. Nitrate isotopic ratios $\delta^{18}O_{NO3}$, $\delta^{15}N_{NO3}$) also showed the consistent seasonality, albeit in opposite direction as nitrate concentration values. $\delta^{18}O_{NO3}$ and $\delta^{15}N_{NO3}$ data were minima values in winter and early spring, and equaled approximately 0‰ and 6‰, respectively. $\delta^{18}O_{NO3}$ and $\delta^{15}N_{NO_3}$ peaks occurred in late summer and fall, and equaled approximately 4‰ and 9‰, respectively. The nitrate isotopic ratio seasonality is shown to coincide with nitrate concentration seasonality, albeit local extrema are in the opposite directions. Nitrate measurements from the two different seasons were statistically significant including $\delta^{15}N_{NO3}$ (p-value $<1 \times 10^{-4}$), $\delta^{18}O_{NO3}$ (p-value $<1 \times 10^{-4}$) and nitrate concentration (p-value $<1 \times 10^{-6}$) based on two-tailed statistical t-tests from samples from the winter-

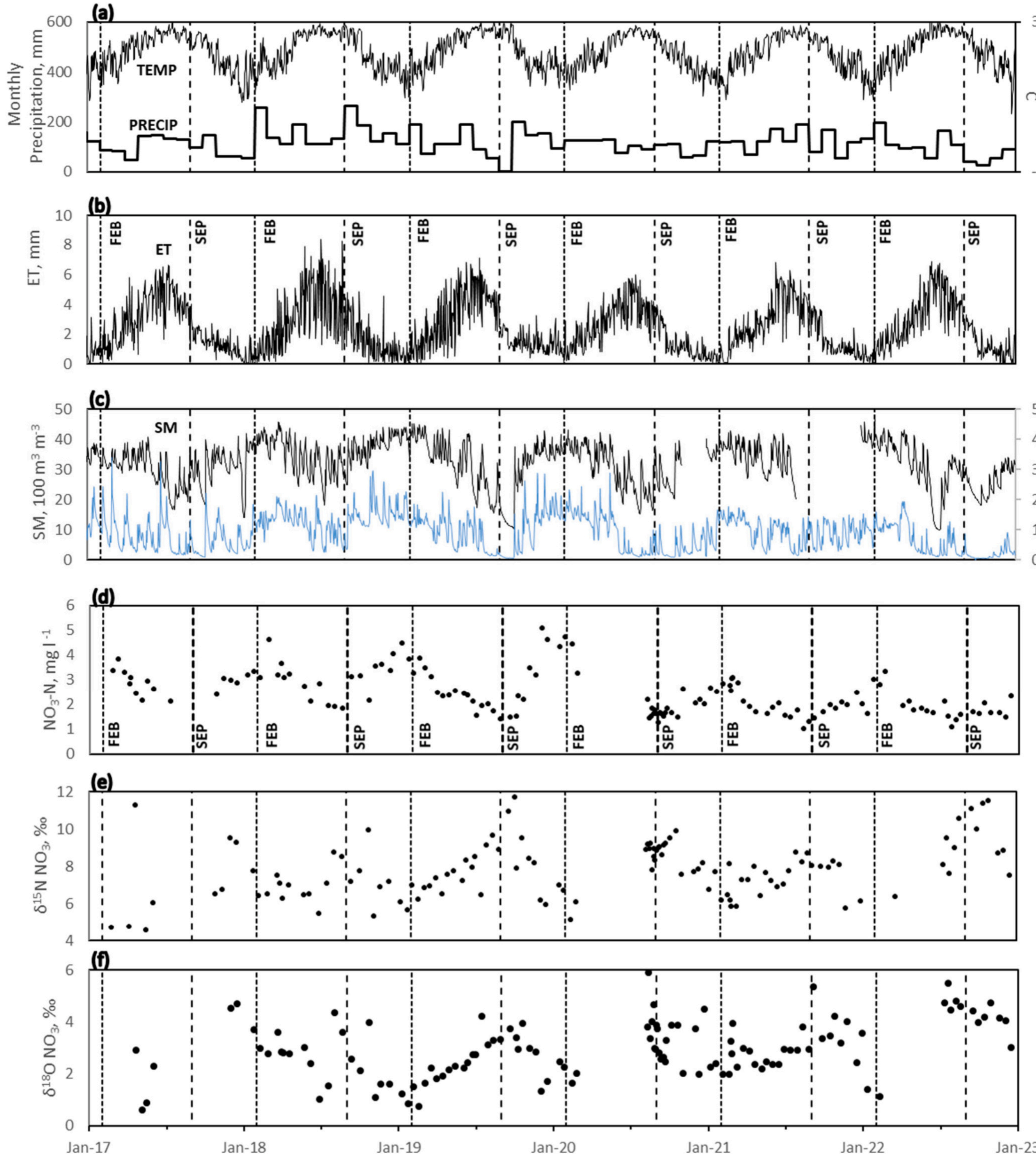

Fig. 4. Measurements results for climate, water flux, nitrate concentration, and isotopic ratios (δ^{15}N and δ^{18}O) of nitrate analyzed for the six year study period. Nitrate concentration and isotopic ratios were data collected weekly to biweekly as discrete (point) samples.

spring versus fall seasons (Hines et al., 2008).

Seasonality of nitrate isotopic ratios and nitrate concentration across the six years duration clearly align with the wetter and drier soil conditions and groundwater regimes (see Fig. 5a). Wetter soil conditions identified with soil moisture in winter and early spring, and on average centered in February, clearly coincide with transport of water with high nitrate concentration and low values for $\delta^{18}O_{NO3}$ and $\delta^{15}N_{NO3}$. Drier soil conditions in late summer and early fall, and on average centered in September, clearly coincide with low nitrate concentration in water and relatively high values for $\delta^{18}O_{NO3}$ and $\delta^{15}N_{NO3}$.

Nitrate isotopic ratio measurements agree with sourcing of nitrate from soil nitrogen and align with nitrate transformation resulting from denitrification (Fig. 6). The isotope data clearly fall within the source box for soil-derived nitrogen, when considering the ranges typically reported (Kendall and Aravena, 2000). Isotopic ratios of the dataset generally fall within the bounds of the 1:1 and 2:1 lines reported in the literature (Kendall and Aravena, 2000) for impacts of denitrification on nitrate isotopic ratio values. The regression slope of the isotopic ratio

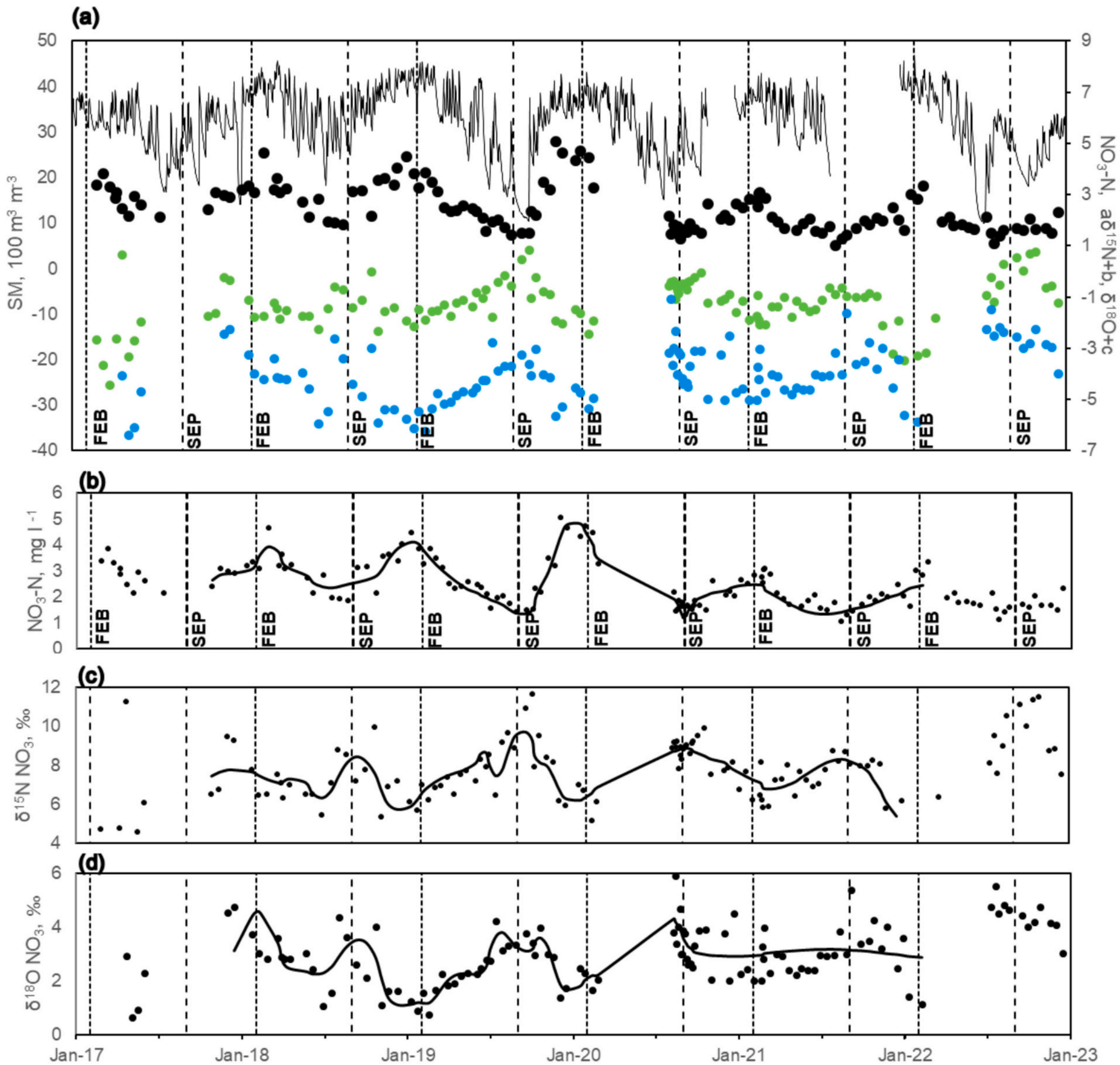

Fig. 5. Visualization and analyses of nitrogen concentration and isotopic ratios of nitrate: (a) Visualization of water regime impacts on nitrate concentration and nitrate isotopic ratios over the six year study. Soil moisture (top black line) is visually correlated with nitrate concentration (black circles); and visually inversely correlated with δ^{15}N (green circles) and δ^{18}O (blue circles) of nitrate. δ^{15}N and δ^{18}O are plotted and δ^{15}N is scaled for visualization purposes, such that the isotopic ratios are plotted as: aδ^{15}N + b and δ^{18}O + c, where a = 0.5, b = −5, and c = −7. Empirical mode decomposition results with significant intrinsic mode functions (IMFs) for (b) nitrate concentration and (c-d) stable isotopic ratios.

dataset equaled 1.5:1, which clearly falls between the denitrification line range (see equation in Fig. 8a).

Taken together, seasonality of nitrate isotopic ratio results ($\delta^{18}O_{NO3}$ and $\delta^{15}N_{NO3}$) is attributed to leaching of nitrate derived from soils during the wetter regime and leaching of nitrate transformed via denitrification during the drier regime. Previous research corroborates the soil origin of nitrate as dominating in this inner bluegrass karst region of central Kentucky, USA, and groundwater modelling results for this basin by Husic et al. (2019) showed that approximately 90% of nitrate transported to Royal Spring originated from soils. Prior research also shows relatively fast groundwater transport from soils to the Royal Spring, which corroborates the inference that nitrate leaching processes are coincident with the groundwater regime. For example, Husic et al. (2019) showed a mean 27 day time of lag of flowrate at Royal Spring relative to rainfall. The time lag of water transported at Royal Spring relative to precipitation fell on the soils will be lower during wet conditions for this basin (Paylor and Currens, 2004), which agrees with the close resemblance of soil moisture fluctuations and Royal Spring flowrate in Fig. 4c. The potential for fast groundwater transport of nitrate leached from soils to the spring outlet is attributed to the mature karst development of the basin, including presence of a well-defined epikarst, fracture network, and phreatic cave that transport water to Royal Spring (Thrailkill et al., 1991; Paylor and Currens, 2004; Al Aamery et al., 2021).

The fast transport from soils to Royal Spring allows discussion of the soil processes dominating each season that likely control the nitrate isotopic ratios in the karst basin, including soil nitrogen mineralization and denitrification (see seasonal concept model in Fig. 7). Relatively newly mineralized nitrate from soil organic nitrogen likely controls $\delta^{18}O_{NO3}$ and$\delta^{15}N_{NO3}$ during the wetter winter season (Fig. 7, panel a). In

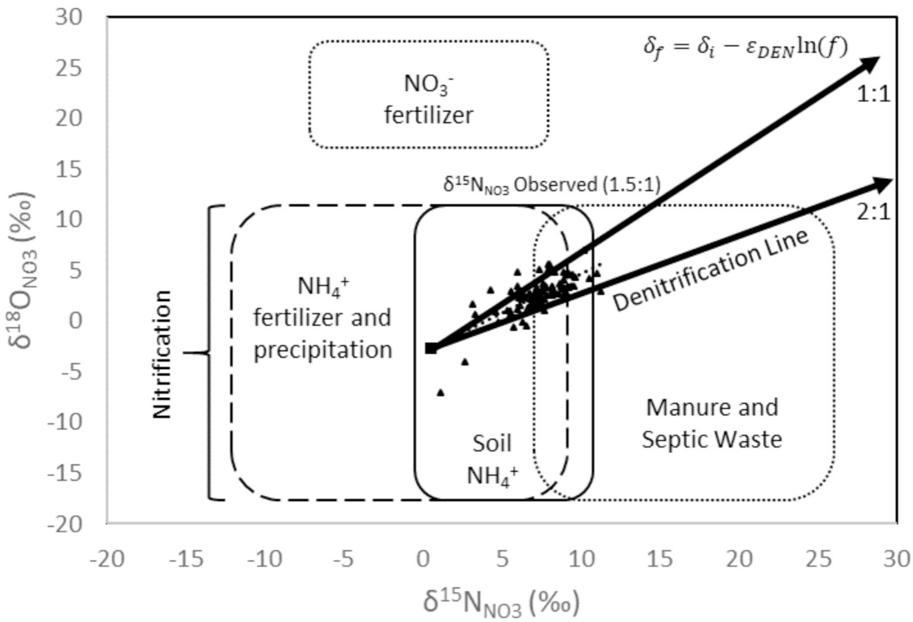

Fig. 6. Controls on δ^{15}N and δ^{18}O of nitrate plotted with the data from this study, in which the soil nitrogen source is identified as prominent with the influence from denitrification. The two arrows indicate typical expected slopes for data resulting from denitrification of nitrate with initial $\delta^{15}N = 0‰$ and $\delta^{18}O = -2‰$. The typical ranges of $\delta^{18}O_{NO3}$ values produced by nitrification of ammonium and organic matter denoted by "nitrification" as presented by Kendall and Aravena (2000). Nitrate samples collected for this study denoted as triangle points, and follow a similar slope (1.5:1) as expected values presented in literature.

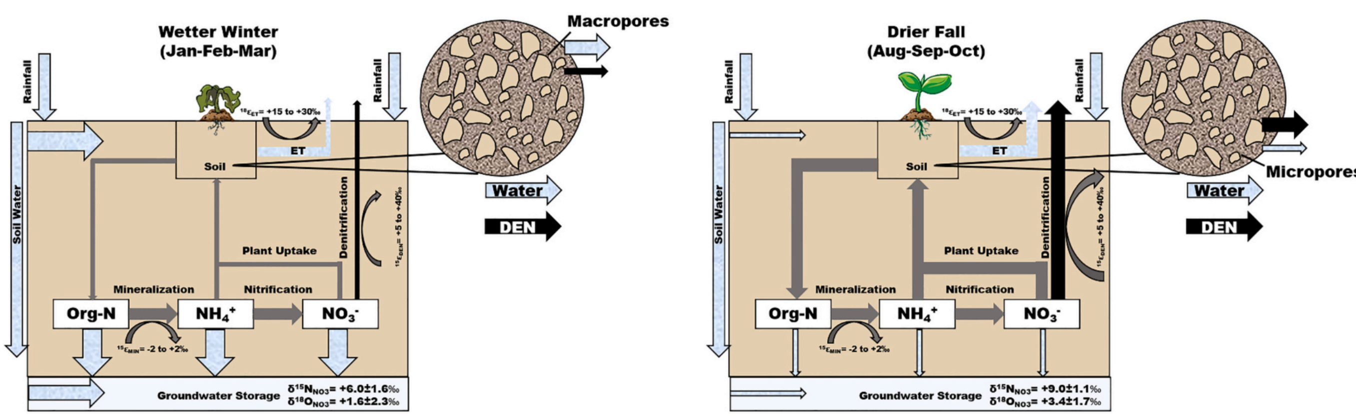

Fig. 7. Concept model of controls on nitrate isotopic ratios during the wetter winter season and drier fall season in the inner bluegrass region.
During the dormant winter months, plant inactivity and low evapotranspiration leads to leaching of mineralized soil nitrate with nitrate isotopic ratios close to that of soil nitrogen. During drier fall months, biological activity results in enhanced nutrient retention by plants and microbes, reduced nitrate mobility, and opportunity for denitrification, such that leached nitrate has isotopic ratios reflecting denitrification.

agricultural soils, such as the hay and pasture of this system, turnover of plant residue can lead to buildup of mineralized nitrate in winter that is leached during hydrologic events (Di and Cameron, 2002). Plants are dormant during the winter season in this region of the United States, and therefore nitrate is leached through the soil rather than being assimilated by plants (Savard et al., 2010; Gerlitz et al., 2023). Transformation of mineralized nitrate, via denitrification, is likely less pronounced given the low residence time of water in the soil system during this wetter regime. Our inference that $\delta^{18}O_{NO3}$ and $\delta^{15}N_{NO_3}$ are controlled by mineralized nitrate from soil organic nitrogen is corroborated by the range of nitrogen isotopic ratios of nitrate relative to soils for the region. $\delta^{15}N_{Soil}$ collected and analyzed as part of this study equaled a range from 4.2‰ to 7.0‰. $\delta^{15}N_{NO3}$ from the wetter winter season equaled a standard error range from 4.4‰ to 7.6‰—showing excellent agreement with that of soil—with an overall nitrate range of all winter data from approximately 1 to 8‰. The very close agreement between δ^{15}N of nitrate in the winter and δ^{15}N of soils supports the soil origin in the wetter winter season.

Nitrate from soil organic nitrogen that is subsequently affected by partial denitrification likely controls $\delta^{18}O_{NO3}$ and $\delta^{15}N_{NO3}$ during the drier fall season (Fig. 7, panel b). Water and nitrate residence time likely increases in soils in the drier fall season, as evidenced by the reduced water flowrate from the system while still substantial soil moisture present during the fall months (Fig. 4c). These conditions are more suitable denitrifying sites as the residence time in pore spaces, i.e., microsites, provides opportunities for oxygen-limited pockets where microbes may reduce stored nitrate (Wang et al., 2020; Schlüter et al., 2024). Denitrification is well established to increase δ^{18}O and δ^{15}N of the residual nitrate pool Böttcher et al., 1990; Panno et al., 2006; Chen et al., 2009). The inference of denitrification control on the $\delta^{18}O_{NO3}$ and $\delta^{15}N_{NO3}$ during the drier fall season is corroborated by the comparison with δ^{15}N of soils collected. $\delta^{15}N_{NO3}$ from the drier fall season equaled a standard error range from 7.1‰ to 10.2‰, with an overall range of all fall data from approximately 6 to 12‰. The nitrate isotopic ratios in this season fall well above the range for $\delta^{15}N_{Soil}$ (4.2‰ to 7.0‰), thus providing evidence towards the potential for denitrification.

Our results suggest origin of nitrate in the groundwater basin is primarily from mineralized soil nitrogen that is denitrified along its pathway with the extent varying seasonally, and the location of the most prominent denitrification sites is another point of discussion. Plausible sites where denitrification occurs is within the soil, in anoxic regions of the groundwater basin, and to a lesser degree within deposited sediment in the bed of the cave network (Husic et al., 2019, 2020b). Husic et al. (2019) carried out numerical modelling of nitrogen fate and transport for the soils and groundwater basin, and their results estimated 60% (±35%) of total denitrification in the basin occurred in the soil, 30% (±19%) of denitrification occurred in the saturated groundwater aquifer, and 10% (±7%) of denitrification occurred in the epikarst, dolines and fracture network (Table 6 in Husic et al., 2019). Husic et al. (2020b) used a daily resolution nitrate dataset together with numerical modelling to estimate that denitrification within deposited sediment at the bed of the cave network was on the same order of magnitude as denitrification in soils, river sediment, and lake sediment (Table 2 in Husic et al., 2020b). However, we expect the spatial extent of cave bed sediment deposits to be relatively small in comparison to the spatial extent of the soils and the saturated portion of the aquifer. We do not expect denitrification to be pronounced in the moving water of the primary cave system, as groundwater in the cave is primarily oxic. The dissolved oxygen concentration in water of the cave system varies from approximately 4 to 10 mg l^{-1} (Supplemental Fig. S1). The cave system provides the outlet of the 58 km^2 karst groundwater basin, and the cave drains water originating from the groundwater transfer via fractures, conduits, and the rock matrix and from sinking surface streams via the more than 50 swallets in Cane Run creek (Thrailkill et al., 1991; Paylor and Currens, 2004; Bettel et al., 2022). Based on the dissolved oxygen dataset, we suggest the sinking streams provide one source to help oxygenate the cave. Dissolved oxygen concentration in water of the cave system is approximately 2 to 4 mg l^{-1} lower than dissolved oxygen concentration of a nearby surface stream, but at the same time the time distribution trend of oxygen in the cave water follows closely with that of the surface stream (Fig. S1). The data results support oxic conditions of the cave originating from the sinking streams but also mixing with groundwater with a lower dissolved oxygen concentration. Taken together, knowledge of the basin suggests denitrification likely occurs in soils, anoxic regions of the groundwater basin, and to a lesser degree within deposited cave bed sediment, and future research to elucidate the location of the most prominent denitrification sites provides an area of further research for this basin.

Our results provide evidence of a dominant soil origin for nitrate transported in the groundwater basin, with the nitrate isotopic ratio providing evidence of denitrification occurring for nitrate either in the soils or saturated aquifer. We provide further evidence for nitrate isotope ratio ability to reflect denitrification in the study system by illustrating how $\delta^{15}N$ of nitrate would be expected to change during isotope fractionation following the assumptions of Rayleigh modelling. We considered initial conditions for $\delta^{15}N$ of nitrate to equal $\delta^{15}N$ of soil nitrogen in the isotope fractionation model (see Fig. S2) because enrichment associated with isotope fractionation during soil mineralization to nitrate is reported to be small. For example, the isotopic ratio enrichment was reported for carbon and nitrogen isotopic ratios during organic matter degradation more generally, e.g., plant, litter, soil organic matter, sediment cores, and these studies find isotopic enrichment is generally ±2‰ (Mariotti et al., 1981; Heaton, 1986; Möbius, 2013). Results of the isotope fractionation modelling during denitrification via the Rayleigh model in Fig. S2 show that the residual nitrate $\delta^{15}N$ increases as a function of the fraction of nitrate remaining after denitrification. We parameterized isotopic enrichment during denitrification to be consistent with the range reported for similar media (i.e., ε was varied from 0 to 20‰ based on analyses of results from Granger et al., 2008, Knöller et al., 2011, Wunderlich et al., 2012, and Margalef-Marti et al., 2025). Results of modelling suggest that as little as 20% of the original nitrate would need to be denitrified to reflect the approximately 10‰ results measured for $\delta^{15}N$ of nitrate in the late summer and early fall period; and low estimates for ε result in 50% of the original nitrate undergoing denitrification to produce the 10‰ results (see black dash lines indicating the fraction remaining for each enrichment factor to reach 10‰ in Fig. S2). While denitrification rates vary widely in soils (Seitzinger et al., 2006; Jahangir et al., 2012; Anderson et al., 2014), the results are well in line with estimates and in particular saturated soils were shown to denitrify as much as 5% of nitrate per day (Anderson et al., 2014). We qualify that a Rayleigh-like model invokes closed system assumptions, and reliable estimates isotope fractionation as a function of denitrification requires development of a nitrogen cycle model that is coupled to isotope fractionation. Nevertheless, the results illustrate further process-based evidence, consistent with isotopic enrichment reported elsewhere, that denitrification of soil nitrate could reasonably lead to measurements for $\delta^{15}N$ of nitrate reported in this study.

One potential alternative explanation of results occurs in several fall seasons and is attributed to a secondary source of nitrate from mineralized animal manure. Analyses of the difference between $\delta^{15}N_{NO3}$ and $\delta^{18}O_{NO3}$ shows that several occurrences provide data where the difference is greater than approximately 6‰ (see upper error line in Fig. 5b). We expect increases in $\delta^{15}N_{NO3}$ with no changes to $\delta^{18}O_{NO3}$ could occur when nitrate from mineralized manure is assimilated to the nitrate pool because the nitrogen isotopic ratio of manure is higher than that of soil nitrogen while the oxygen isotopic ratio is consistent with that of soil nitrogen (Kendall, 1998; Panno et al., 2006). The potential for manure to influence the nitrate isotopic ratios is plausible, because there is a high occurrence of equine events at the Kentucky Horse Park during late summer and early fall. This is somewhat corroborated by the results of Currens et al. (2015), where the authors showed correlation of fecal counts in cave water and equine events in late summer and fall periods. Hydrology of this time period includes dry conditions, and runoff during convective events potentially transports animal manure to the stream network that may transport to the cave system via sinking streams. Residence of nitrate-derived manure in the cave system during the low flow fall periods shows the potential for nitrogen mineralization, thereby shifting the $\delta^{15}N_{NO3}$ measured at Royal Spring. In this manner, these results suggest the potential for some influence from the animal manure end member on the isotopic ratio results, albeit the influence is expected to fairly limited relative to the soil nitrogen origin.

4.2. Empirical mode decomposition of $\delta^{15}N$ and $\delta^{18}O$ for nitrate

Empirical mode decomposition of the nitrogen and oxygen isotopic ratio and elemental measurements of nitrate as nitrogen showed statistically significant long term and seasonal variation of data (Fig. 5). Empirical model decomposition showed significant IMFs with nearly one year sinuosity for all parameters that allowed separation of the isotopic ratio and elemental seasonal variation (Fig. 5 b-d). $\delta^{15}N$ and $\delta^{18}O$ showed local minima in winter in most years while NO_3-N showed winter local maxima in most years. Please note that $\delta^{15}N$ and $\delta^{18}O$ are scaled values in the Fig. 5a, which provides easier comparison with the concentration data and hydrologic measurements. Nitrate isotopic ratios showed local maxima in late summer and fall while NO_3-N showed local minima in fall most years.

The significant IMFs decomposed from the isotopic ratio data showed pronounced changes in seasonally from year to year. At times, large shifts from winter minima to fall maxima exceeded 4‰ for both $\delta^{15}N$ and $\delta^{18}O$ (Fig. 5 c-d). Elemental shifts are also substantial and were as high as 3.5 mg N l^{-1} for nitrate (Fig. 5b). On average annual seasonal shifts for $\delta^{15}N$, $\delta^{18}O$, and NO_3-N were 3.2‰, 2.8‰, and 2.5 mg N l^{-1}, respectively (Fig. 5b-d).

The statistical results from the empirical mode decomposition provide quantitative results to the visualization in Section 4.1. In this manner, the statistically significant longer duration IMFs clearly quantify the seasonality suggested in Section 4.1.

4.3. Seasonality of the denitrification line

We considered the seasonality of the denitrification line, in the context of the present study in the inner bluegrass karst region. The nitrate isotopic ratio biplot shows the previously mentioned denitrification line for all data collected (Fig. 8a). Also, we plot the denitrification line for data collected from winter and fall (blue squares and red triangles, respectively, in Fig. 8b), which is likely controlled by mineralized soil nitrogen and nitrate that undergoes denitrification, respectively. The biplot is consistent with our time series results and shows that δ^{15}N of nitrate in winter (5.95 ± 1.60‰) is close to that of soil (5.60 ± 1.55‰); and δ^{15}N of nitrate in fall (8.65 ± 1.6‰) reflects enrichment of N-15 and is consistent with trends in denitrification. The clear separation of the dataset between winter and fall dataset supports the importance of denitrification on nitrate isotopic ratios in the fall season.

Results suggest seasonality of the denitrification line, especially for studies such as the inner bluegrass where a single nitrate source (i.e., mineralized soil nitrogen) is transported in the karst groundwater system. The six-year long time series dataset is one contribution of this paper that is fairly unique to the published literature of nitrate isotopic ratios from river water or groundwater. The long dataset allows

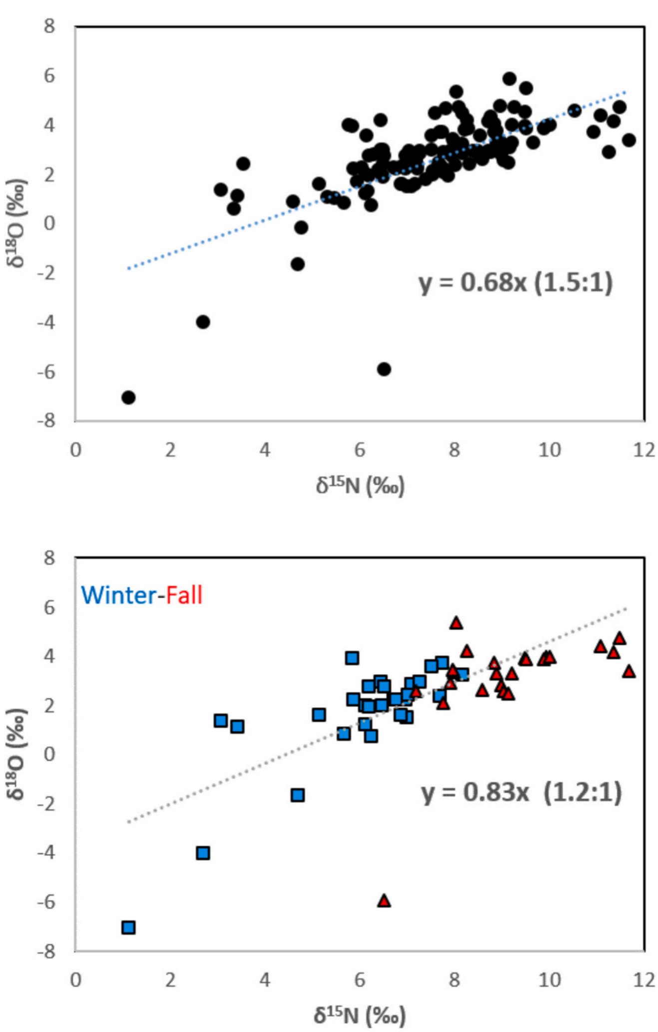

Fig. 8. Range of δ^{15}N and δ^{18}O values for nitrate collected and observed. Denitrifying bacteria cause the N and O isotopes to fractionate, increasing the δ^{15}N and δ^{18}O values of the remaining nitrate.

evidence that supports the seasonally controlled denitrification line.

4.4. Seasonality of the Rayleigh diagram and isotope enrichment

We also considered seasonality of the Rayleigh diagram and isotopic enrichment of both ^{15}N and ^{18}O during fractionation. The Rayleigh diagram is less reported, in comparison with the denitrification line, in studies that use nitrate isotopic ratio for considering nitrate source and transformation in surface water and groundwater systems. For example, 24 out of the 28 journal papers in Table 1 publish or refer to the denitrification line for analyzing their dataset. And, only 12 out of 28 journal papers in Table 1 publish a Rayleigh diagram. The typically reported Rayleigh diagram plots δ^{15}N of nitrate as the dependent variable (e.g., y-axis) as a function of the logarithm of nitrate concentration as the independent variable (e.g., x-axis). This slope is not the isotope fractionation factor or enrichment, but is rather the gradient of the δ^{15}N versus ln(NO_3) and is customarily with that reported in other papers (Widory et al., 2004; Griffiths et al., 2016; Biddau et al., 2019; Bourke et al., 2019) showing the linear trend increasing δ^{15}N with the decreasing logarithm of the NO_3^- concentrations. The units for these plots are analogous to Rayleigh enrichment factor (‰). The negative sloped Rayleigh diagram results suggests denitrification because sources of nitrogen will plot in the lower right, while sources that undergo denitrification will plot in the upper left indicating lower nitrate concentration (i.e., from loss of nitrate via denitrification) and corresponding higher values of δ^{15}N of nitrate since N-14 is preferentially removed during denitrification.

When plotting the Rayleigh diagram for the inner bluegrass dataset,

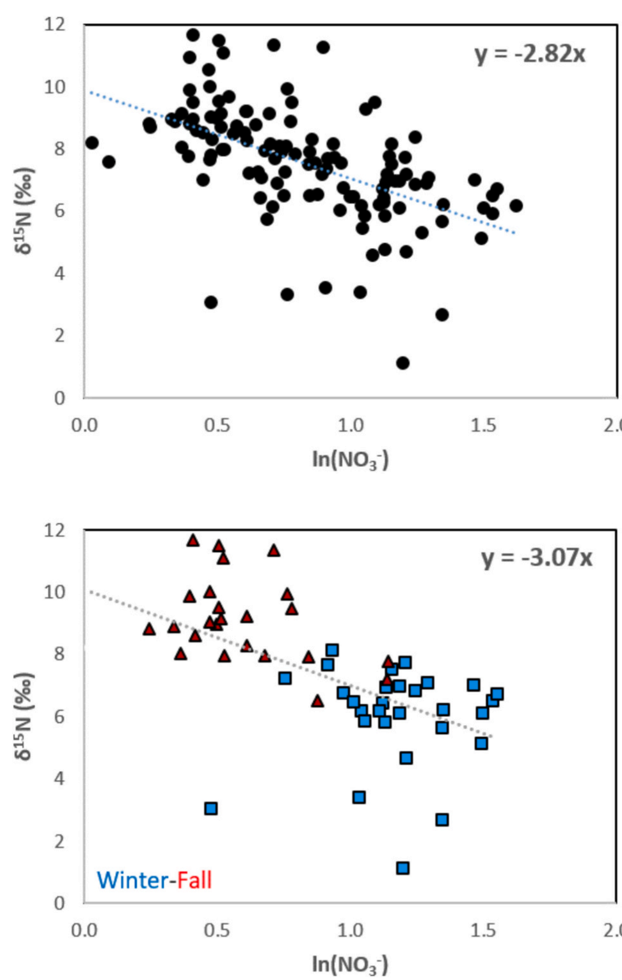

Fig. 9. Rayleigh plots showing δ^{15}N and ln(NO3) for nitrate observed.

the data clearly indicates the seasonality of the Rayleigh diagram (Fig. 9). Lower relative values for δ^{15}N of nitrate collected in winter months clearly align with higher nitrate concentration in water, and higher δ^{15}N of nitrate in summer aligns with lower nitrate concentration in water indicating changes of the measurements due to denitrification.

The Rayleigh diagram is useful to suggest the occurrence of denitrification in the system; however, the closed system assumption of the Rayleigh model limits its ability, and we argue should be approached with caution. The dominance of the single soil mineralization source is reasonable, and winter mineralized nitrate concentrated in water provides a reasonable estimate for initial conditions. And, during the growing season, nitrogen isotopic fractionation of nitrate during plant uptake is likely small (Kohl and Shearer, 1980; Yoneyama et al., 1991; Kendall and Aravena, 2000). However, in absence of a full terrestrial ecosystem model, an estimate of the amount of nitrate uptake by plants relative to the fraction lost via denitrification is not possible with the given dataset. Further work might leverage the existing dataset with a terrestrial ecosystem model for estimating the isotopic enrichment. Regardless, the results Fig. 9 further justify the importance of denitrification in controlling the results in the inner bluegrass karst region.

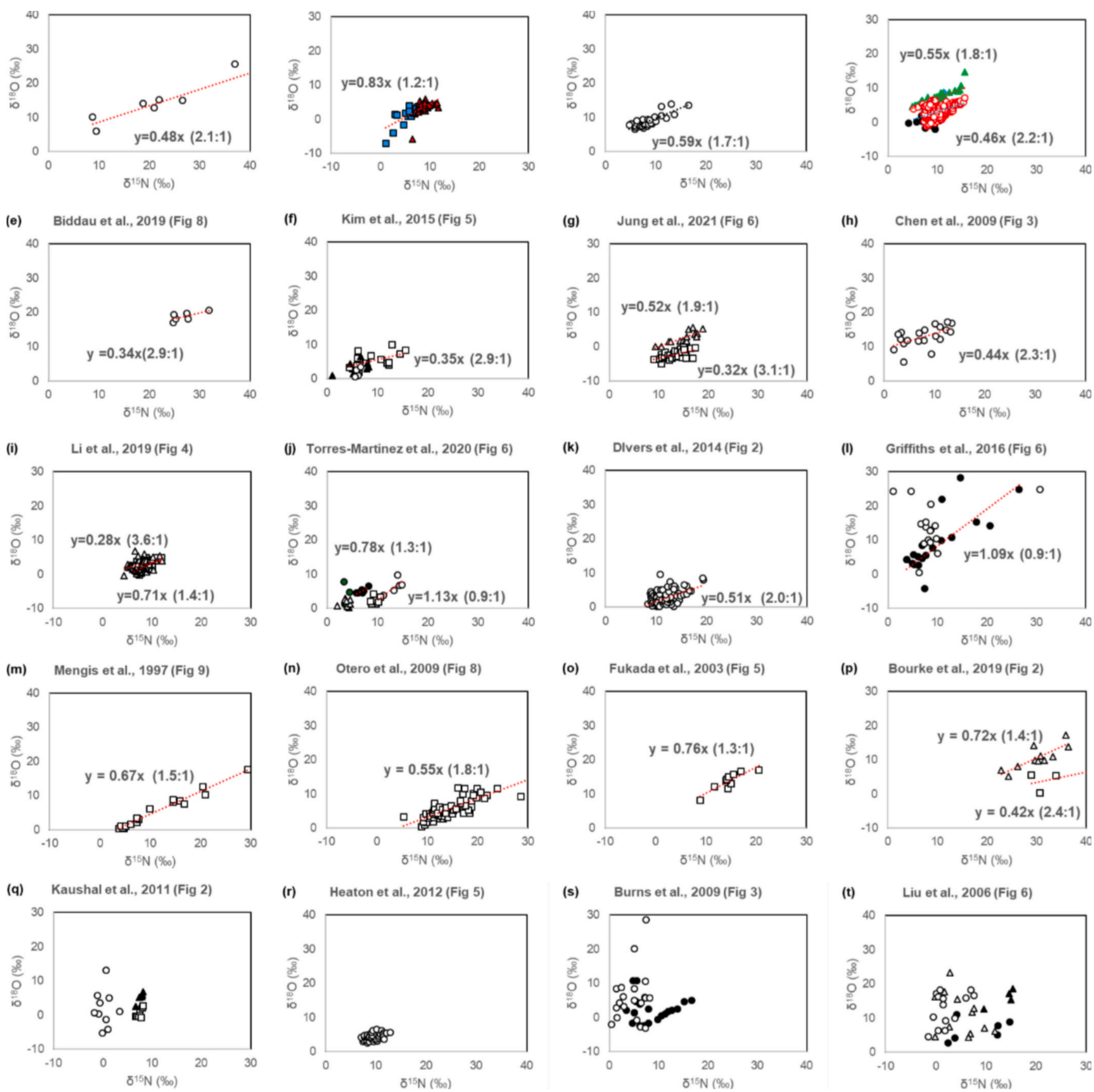

Fig. 10. Meta analyses of nitrate isotope data presented in surface and groundwater studies. Each study is qualified based on importance of denitrification fractionation and end member mixing nitrogen sources to explain nitrate isotope variance through dataset. Isotope bi-plots (a-t) are ranked from dependence on denitrification dominated (top left) to end member-mixing controlling (bottom right).

4.5. Meta analysis of controls on the nitrate isotopic ratios in surface and groundwater studies

Our meta-analysis of the previously reported $\delta^{18}O_{NO3}$ and $\delta^{15}N_{NO3}$ in the journal papers of Table 1 suggests a continuum from denitrification controlled to end member mixing controlled conditions for water collected from groundwater and surface water basins. Fig. 10 (top panel) organizes the meta-analysis such that the denitrification-dominated systems fall at the far left and source end member mixing dominance falls to the far right. As noticed in Fig. 10, plots a through x, the strength of the denitrification line in $\delta^{18}O_{NO3}$ and $\delta^{15}N_{NO3}$ bi-plots correlates well visually with the plotting of the study's prominent control in the top panel Fig. 10. Studies with pronounced denitrification control show a clear denitrification line when comparing the left side of Fig. 10 top panel with Fig. 10 a through p. Studies with dominance of end member mixing show a lack of clear denitrification line in nitrate isotope biplots, shown on the right side of Fig. 10 top panel with Fig. 10 q through x.

Denitrification controlled studies from the meta-analyses (far left panel in the top of Fig. 10, i.e., Böttcher et al., 1990; Riddle et al., 2025-this study; Granger et al., 2008; Panno et al., 2006; Fukada et al., 2003; Otero et al., 2009; Lin et al., 2019) show dominance of the denitrification line, and $\delta^{18}O_{NO3}$ and $\delta^{15}N_{NO3}$ may be most useful to help parameterize denitrification in surface and groundwater studies of nitrate fate. The Granger et al. (2008) study is included as a benchmark for the denitrification dominance, and this was a laboratory investigation of isotope fractionation for cultures of denitrifying bacteria. Böttcher quantified fractionation during denitrification in an aquifer groundwater system, and their consistent linear relationship for $\delta^{15}N$:$\delta^{18}O$ suggests a single dominant source that exhibits denitrification (Fig. 10a). The present study (Riddle et al. in Fig. 10b) exhibits denitrification of the dominant soil nitrogen source and clearly visible denitrification line. Panno et al. (2006) indicated denitrification occurring in a single soil source prior to soil water discharging to the Mississippi River, and they present a linear relationship for $\delta^{15}N$:$\delta^{18}O$ (Fig. 10c).

Studies that indicate denitrification dominance with some influence of multiple end member sources (second panel from left at the top of Fig. 10, i.e., Mengis et al., 1999; Chen et al., 2009; Kim et al., 2015; Bourke et al., 2019; Lin et al., 2019; Biddau et al., 2019; Li et al., 2019; Jung et al., 2021) provide results that reflect the denitrification line, though overlapping end-member sources must be considered. Lin et al. (2019) reported two principal nitrate sources using the isotopic ratios, including wastewater treatment plant (WTP) effluent and agricultural groundwater, from a river network; and the isotopic ratios of each source exhibited their own similarly sloped denitrification lines (Fig. 10d). Biddau et al. (2019) reported the isotopic ratios of nitrate with evidence of denitrification in a shallow groundwater system with sources attributed to manure and natural attenuation (Fig. 10e). Chen et al. (2009) reported results that indicate the primary source of nitrate was nitrification of fertilizer, manure and sewage with seasonal influence observed in denitrification (Fig. 10h). Divers et al. (2014) quantified the sources of nitrate in an urban stream, specifically focusing on the contributions from sewage and atmospherically derived nitrogen; and observed the influence of denitrification via the denitrification line (Fig. 10k). Griffiths et al. (2016) presented nitrate isotope data of groundwater that suggested denitrification and other processes and mixing of sources affect nitrate concentrations, albeit showing the denitrification process to some extent via the denitrification line (Fig. 10l). These studies provide examples of systems that show the strong influence of the denitrification process, via the denitrification line, on nitrate in groundwater or stream water that include nitrate originating from more than one source.

A number of studies indicate isotopic ratios showing the dominant control of end member mixing of multiple sources with modest to low control from denitrification (second panel from right at the top of Fig. 10, i.e., Heaton et al., 2012; Burns et al., 2009; Liu et al., 2006; Kaushal et al., 2011). Heaton et al. (2012) identified fertilizers, manure, sewage systems, and soil as the main potential sources of nitrate impacting isotopic ratios, with evidence for denitrification fractionation in only one sample throughout the study duration (Fig. 10r). Burns et al. (2009) identified seasonal variation of sources of nitrate from atmospheric nitrate in forested watersheds and animal waste in agricultural land; and their results showed weak influence of denitrification via the denitrification line (Fig. 10s). Liu et al. (2006) reported nitrate sourcing controlled by fertilizers, soil organic nitrogen, and livestock waste, and recognized denitrification processes in isotopic ratios but do not quantify due to strong source mixing dominating the signal (Fig. 10t). These studies provide examples of systems that show nitrate isotope variance due to dominance of source mixing with denitrification as less important, suggesting that minimal denitrification affects nitrate source estimates in these systems.

Studies using end member mixing with little to no isotope fractionation during denitrification from the analyses (far right panel in top of Fig. 10, i.e., Kendall et al., 1995; Burns and Kendall, 2002; Campbell et al., 2002: Moore et al., 2006; Pardo et al., 2004; Savard et al., 2010) may be helpful to investigate end member mixing, although will be limited for assessing denitrification. Pardo et al. (2004) reported nitrate isotope data of stream water to differentiate between atmospheric deposition and microbial nitrification, and the results showed little evidence of the denitrification line (Fig. 10u). Moore et al. (2006) identified animal waste and synthetic fertilizer dominate nitrate sources and indicated denitrification is not important to controlling nitrate isotope variance. Savard et al. (2010) qualified soil organic matter, chemical fertilizers, and manure as the main nitrogen sources with groundwater being sufficiently oxygenated to limit denitrification processes (Fig. 10v). Kendall et al. (1995) found atmospheric deposition and soil-derived nitrate dominate sources with seasonal dependence during snowmelt periods and little denitrification influence (Fig. 10w). These studies provide examples of systems that differentiate nitrogen sources and allocate contributions with little to no effect from nitrogen isotope fractionation processes during denitrification.

As one discussion point, the meta-analysis results that classifies nitrate isotopic ratio studies across the continuum of denitrification to end-member dominated reveals the karst groundwater system behavior in the context of prior studies reported worldwide. Our isotopic ratio results from the karst system agree well with the laboratory study of denitrification and studies showing denitrification from a single nitrate source (Böttcher et al., 1990; Granger et al., 2008; Panno et al., 2006; Lin et al., 2019). This result is consistent with our assumption of the fast groundwater transit in karst. The Royal Spring groundwater basin is dominated by soil water drainage and hence soil nitrate origin, and the fast transit of nitrate provides an integrated signal from the single dominant source. The results support our hypothesis that the relatively long 6-years duration dataset from the karst groundwater provides measurements from karst sources that exhibit seasonality; and in the present study the seasonality is attributed to control by denitrification of nitrate mineralized from soils.

A second discussion point worth future consideration is the dependence of the denitrification line slope on the location of denitrification, such as denitrification occurring primarily in anoxic groundwater as compared to denitrification occurring in soils. We add discussion to this question by considering the conditions of the studies in which denitrification was expected to dominate the denitrification line (see Fig. 10), including the studies by Böttcher et al. (1990), this study (Riddle et al. in Fig. 10), Panno et al. (2006), and Lin et al. (2019). Böttcher et al. (1990) measured denitrified nitrate in an unconfined shallow groundwater where presence of anoxic conditions were prevalent and showed the denitrification line slope of 2.1:1. compared against systems with surface and groundwater interactions. This study and Panno et al. (2006) present results in mixed systems where soil denitrification is expected to be high and anoxic groundwater may also be a location of denitrification, yet aerobic surface water is expected to be prevalent in portions of the groundwater system (i.e., the cave system in this study); and these results show denitrification line ranging from 1:1 to 1.7:1. The results anecdotally suggest a dependence on the primary location of denitrification occurrence. This idea of oxygen-limited aquifers showing denitrification lines exceeding 2:1 slopes is further displayed in an Illinois River Basin study by Lin et al. (2019), where seasonal patterns of a mixed surface water and groundwater basin show potential to influence slopes. Lin et al. (2019) showed that samples collected in Spring months show a slope of 1.8:1 and a steeper slope in summer/fall months equal to 2.2:1 reflect oxygen-depleted groundwater dominating fluxes in drier soil moisture conditions. Work presented in laboratory cultures in Granger et al. (2008) are not able to support relationships between oxygen availability and denitrification of nitrogen and oxygen isotopic fractionation because oxygen is one variable over which the study had limited control during culturing. However, the location of denitrification provides an interesting question to explore in future laboratory studies on cultures of denitrifying bacteria influence in differing environment conditions (e.g., redox/oxygen concentrations).

As a third point, the meta-analysis results allow discussing tracer-aided numerical modelling with nitrate isotopic ratios, which requires differentiating between transformation processes and nitrogen source mixing, as nitrate isotopic ratio variance is influenced by both factors. Modellers may use the denitrification line results in Fig. 10 as a guide, and we recommend these considerations as indicative of the usefulness of the isotopic ratios in modelling, or lack thereof. Results from the far left panel of Fig. 10 show examples of where the isotopic ratios will likely be most useful for calibrating denitrification rates in tracer-aided numerical modelling. Results from the second from left panel of Fig. 10 show examples of where the potential for helping to calibrate denitrification is possible; however, the researcher should be cautioned that the multiple sources and their denitrification processes should be considered separately in the model. The multiple sources add the potential for equifinality in modelling, and additional information of source allocation, i.e., using on the ground information or other tracers, would improve confidence. Results from the second panel from right and the far right panel of Fig. 10 show that calibration of denitrification processes via the isotopic ratios will likely be limited in numerical modelling, although the isotopic ratios may be useful for end member mixing to some degree. This is especially true for results from the far right panel, and these examples present studies where it is reasonable that the isotopic ratios would be may be useful as conservative tracers driven by hydrology.

5. Conclusion

The conclusion of this study is as follows:

1. Nitrate isotopic ratio measurements, as well as nitrate concentration, showed seasonal patterns across the six-year dataset from the inner bluegrass karst region, Kentucky, USA that were consistent with the wetter and drier soil conditions and groundwater regimes. The seasonality of the isotopic ratios of karst waters, including low $\delta^{18}O_{NO3}$ and $\delta^{15}N_{NO3}$ measurements in wet winters and high $\delta^{18}O_{NO3}$ and $\delta^{15}N_{NO3}$ measurements in dry summer-fall periods, is attributed to nitrate leaching from soils and denitrification occurring in the soil system. The dominance of the soil nitrogen origin for nitrate and the fast transport of water from soils to the karst spring allowed a unique investigation of controls on $\delta^{18}O_{NO3}$ and $\delta^{15}N_{NO3}$.
2. Results show seasonality of denitrification line for the present study, which is attributed to the dominant soil nitrogen source for nitrate and the denitrification control. Results also show seasonality of the Rayleigh diagram for the inner bluegrass dataset: lower values for $\delta^{15}N$ of nitrate collected in winter months clearly align with higher nitrate concentration in water; and higher $\delta^{15}N$ of nitrate in summer aligns with lower nitrate concentration in water indicating changes of the measurements due to denitrification.
3. Meta-analyses results suggest the continuum of studies from $\delta^{18}O_{NO3}$ and $\delta^{15}N_{NO3}$ being controlled by denitrification to control by end-member mixing. Meta-analysis results suggest successful applications of nitrate isotopic ratios in the structure of tracer-aided modelling will likely vary from highly useful to limited. Surface and groundwater basins with a single end member source of nitrate show the potential for calibrating denitrification processes in modelling via the isotopic ratios while studies controlled by relatively fast transport (i.e., low residence time) of multiple nitrate sources will show usefulness of end member mixing analyses using the isotopic ratios as conservative tracers. Researchers should exercise more caution when multiple prominent sources exhibit some denitrification influence, as equifinality will need to be considered in modelling. The meta-analysis herein might provide a guide to considering these conditions in designing future studies.

As one final consideration, future research should focus on collecting long-term time series data to better understand the seasonality of nitrate data and its connection to denitrification processes. Additionally, more studies are needed to explore the nitrate seasonality in karst groundwater systems with agricultural land use, as these systems remain understudied.

CRediT authorship contribution statement

Brenden Riddle: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. **Jimmy Fox:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **William Ford:** Supervision, Formal analysis. **Minjae Kim:** Writing – review & editing, Supervision. **Admin Husic:** Writing – review & editing. **Jason Backus:** Resources, Data curation. **Erik Pollock:** Supervision, Resources, Methodology, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scitotenv.2026.181715.

Data availability

Data will be made available on request.

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